PROGRAMMING FOR PROBLEM SOLVING LAB

I - II Semester: **Common to all branches**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS03	ESC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100

COURSE OBJECTIVES:

1) To be familiarize with flowgorithm to solve simple problems
2) To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
3) To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings, pointers and structures.

COURSE OUTCOMES:

At the end of the course, student will be able to

1) Solve simple mathematical problems using Flowgorithm.
2) Correct syntax errors as reported by the compilers and logical errors encountered at run time.
3) Develop programs by using decision making and looping constructs.
4) Implement real time applications using the concept of array, pointers, functions and structures.
5) Solve real world problems using matrices, searching and sorting.

LIST OF EXPERIMENTS

Week-1 INTRODUCTION TO FLOGORITHM

1. Installation and working of Flowgorithm Software.
2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc.
3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle).
4. Draw a flowchart to find the sum of individual digits of a 3 digit number
5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored.
6. Draw a flowchart to find roots of a quadratic equation.
7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd
8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene

triangle

9. Draw a flowchart to check whether a given number is palindrome or not.

Week-2	**BASIC DATA TYPES**

1) Write a C program to find division of 1 integer 1 float numbers.
2) Write a C program to find division of 2 integer numbers.
3) Write a C program to find average of (1int,1float) numbers.
4) Write a C Program to Swap Numbers without Using Temporary Variable
5) Write a C Program to Swap two Numbers Using Temporary Variable
6) Write a C Write a program to read the values of x, y and z and print the results of the following expressions in one line.
   a) (x+y+z) / (x-y-z)
   b) (x+y+z) / 3
   c) (x+y) * (x-y) * (y-z)

Week-3	**OPERATORS**

1) Write a C program to convert temperature from Fahrenheit to Celsius and vice versa (c=(fh-32)/1.8
2) Write a C program to find area and perimeter of a circle. (area=∏r2 perimeter=2∏r)
3) Write a C program to calculate area and perimeter of a right angled triangle.
   a. (Area=1/2*b*h perimeter=w+h+sqrt(w*w+h*h)
4) Find the sum of natural numbers 1 to n.(read n as input) (Use formula sum=n(n+1)/2
5) Write a C program to calculate Simple interest (SI=PTR/100)
6) Write a C program to calculate area and perimeter of a rectangle. Area=l*b Perimeter=2* (l+b)
7) Write a C program to calculate the value of the third angle of a triangle if two angles are given as input.(a+b+c=180)
8) Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. (Amt=num of units*cost)
9) Write a C Program to calculate area and perimeter of a triangle.
   a. Perimeter=(a+b+c)
   b. s=(a+b+c)/2
   c. Area=sqrt(s*(s-a)(s-b)*(s-c))
10) Write a C program to read five Subject marks and find the average.
11) Write a C program to Calculate Compound interest (CI=p(1+r/100)n

Week-4	**CONDITIONAL STATEMENTS**

1. Write a C program to find largest and smallest of given numbers.
2. Write a C program which takes two integer operands and one operator form the user(+,-,*,/,% use switch)
3. Write a program to compute grade of students using if else adder. The grades are assigned as followed:

   marks<50 F
   50≤marks< 60 C
   60≤marks<70 B
   70≤marks B+
   80≤marks<90 A
   90≤mars≤ 100 A+
4. Write a C program to whether given year leap year or not.
5. Write a C program to find whether given triangle is scalene or isosceles or equilateral.

Week-5	**LOOPING STATEMENTS**

1) Write a C program to find Sum of individual digits of given integer
2) Write a C program to generate first n terms of Fibonacci series
3) Write a C Program to find the Sum of Series SUM=1-x2/2! +x4/4!-x6/6!+x8/8!-x10/10!
4) Write a C program to print the Fibonacci sequence up to given value of n.
5) Write a C program to print the multiplication table to the given value of n.
6) Write a C program to check whether given number is palindrome or not
7) Write a C program to check whether given number is perfect or not

Week-6	**NESTED LOOPING STATEMENTS**

1) Write a C program to generate prime numbers between 1 and n
2) Write a C program to generate Pascal's triangle.
3) Write a C program to generate the following pyramid of numbers.

```
        1
    1   3   1
1   3   5   3   1
```

4) Write a C program to generate the pattern

```
*****
 ****
  ***
   **
    *
```

Week-7	**ARRAYS**

1) Write a C Program to implement following searching methods
   i. Binary Search

ii. Linear Search
2) Write a C program to find largest and smallest number in a list of integers
3) Write a C program
   i. To add two matrices
   ii. To multiply two matrices
4) Write a C program to find Transpose of a given matrix

Week-8	FUNCTIONS

1) Write a C program to find the factorial of a given integer using non recursive functions
2) Write a C program to find GCD of given integers using non recursive functions
3) Write a C Program to find the power of a given number using non recursive functions
4) Write a C program to find sum of natural numbers using non recursive function.
5) Write a C program to reverse a given integer number using non recursive functions
6) Write a C Program to find binary equivalent of a given decimal number using recursive functions.
7) Write a C Program to print Fibonacci sequence using recursive functions.
8) Write a C Program to find LCM of 3 given numbers using recursive functions
9) Write a C program to find the factorial of a given integer using recursive functions
10) Write a C program to print fibonocci series till n terms using recursion.

Week-9	STRINGS

1) Write a C program using to Insert a sub string into a given main string from a given position
2) Write a C program using to Delete n characters from a given position in a string
3) Write a C program to determine if given string is palindrome or not
4) Write C Programs to demonstrate the following string handling functions.
   a. strcat()
   b. strcmp()
   c. strrev()
   d. strcpy()
   e. strlen()
   f. strstr()
   g. strncpy()
   h. strncat()
   i. strncmp()

Week-10	POINTERS

1) Write a C program to read the elements of 1-d array using pointers and print them in

reverse order using pointers.

2) Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference.
3) Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array.
4) Write a C Program to print 2-D array using pointers.

Week-11	STRUCTURES

1) Write a C Program using functions to
    a) Reading a complex number
    b) Writing a complex number
    c) Add two complex numbers
    d) Multiply two complex numbers

   Note: represent complex number using structure
2) Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.

Week-12	FILES

1) Write a C program to read and print the content of a file.
2) Write a C program copy the content of one file to another file
3) Write a C program to merge two file into third file.
4) Write a C Program to find the number of lines in a text file

Text Books:

1) Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014.
2) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13th Edition, 2012.

Reference Books:

1) Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018
2) King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2nd Edition, 2015.
3) Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3rd Edition, 2004.
4) Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1st Edition, 1994.